One bridge.
Four strings.

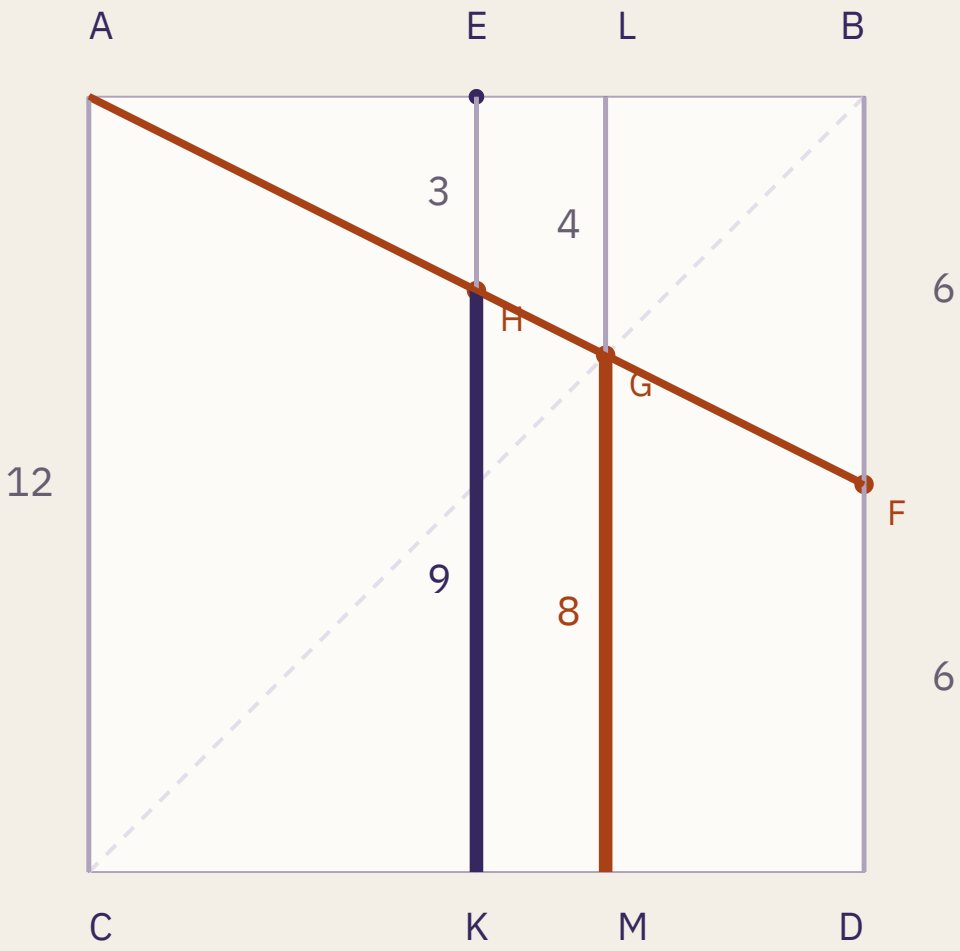
Build the source’s geometry, hear its network of consonances, then retune two inner strings while keeping the end interval fixed. This document retains the experiment captured when you pressed Save.

An ideal geometric and listening experiment after Ptolemy, Harmonics II.2. Uniform strings share the same full-length pitch; a zero-width bridge sets their sounding lengths. Tension, material, bridge height and coupling are held outside this model. Sine tones, absolute reference pitch and the four-string retuning exercise are modern choices. This is not a reconstruction or a measurement of an ancient instrument.

Editable JSON Exact CSV Diagram SVG Retuning WAV Interval WAV

The source board and your bridge

Construction stage 5. Right bridge contact 1/2 of the full height. Full-string reference 220 Hz. The initial square and four string positions follow the source’s first figure; altered contacts keep those strings fixed.



Length units: AC = 12. One shared bridge: AF.

> HK → GM: frequency ratio 9/8 · 203.91 cents. Agrees with the source's 9/8 whole tone.

The interval sound plays the selected first segment, then the second, using their current modeled frequencies. No ancient timbre is inferred.

Sounding lengths on both sides of the bridge · full string = 12

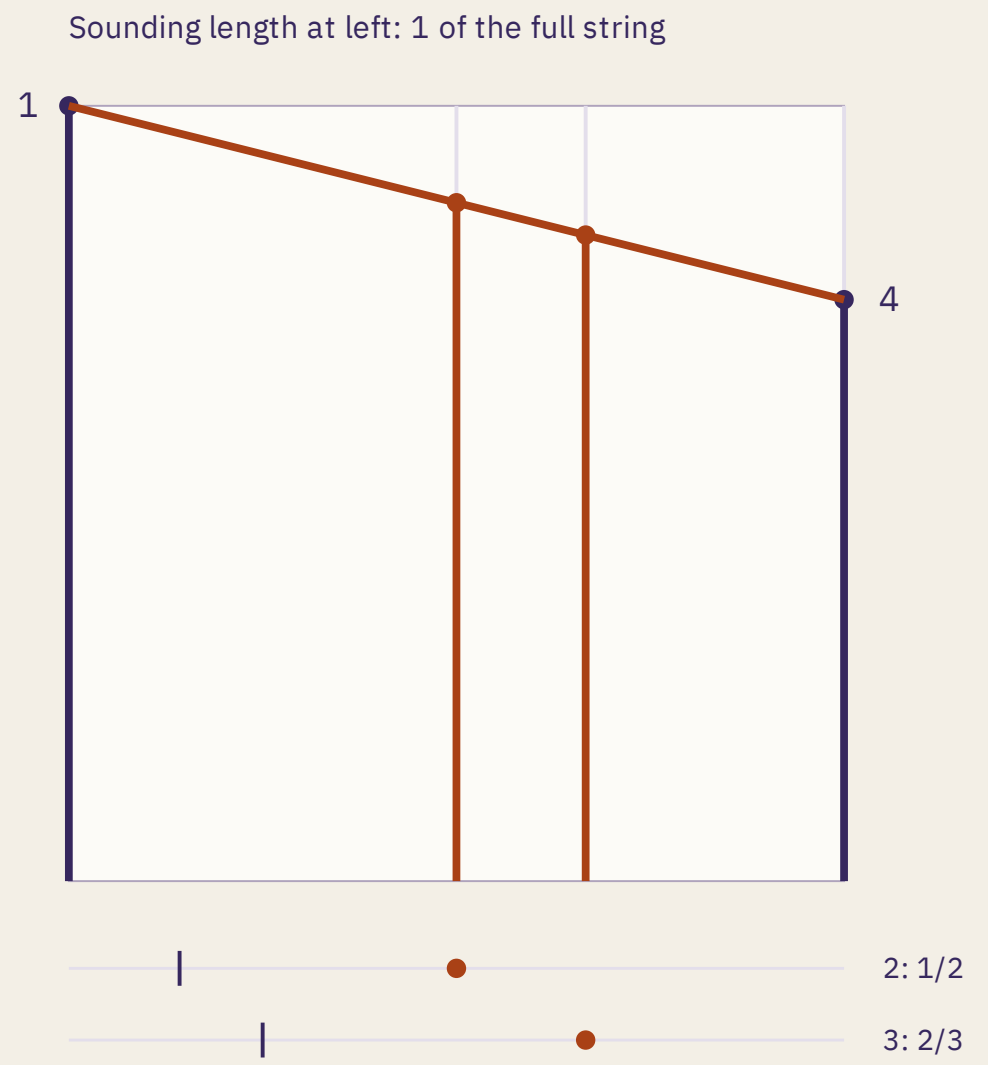
Segment	Length	Pitch / full string	Hz
AC	12	1	220
HK	9	4/3	293.333
GM	8	3/2	330
FD	6	2	440
BF	6	2	440
LG	4	3	660

Segment	Length	Pitch / full string	Hz
EH	3	4	880

Seven nonzero segments have six distinct lengths at the source setting: BF and FD are both 6. The number labels use a full string of length 12, without choosing an ancient unit or a physical size.

Keep the ends; move the inner strings

Target: enharmonic. Inner positions 1/2 and 2/3 of the width. Leftmost sounding length q = 1 of the full string.



2 of 4 target notes agree. The end notes fix the bridge line. Its two inner pitches still depend on where their strings cross it.

Actual and target frequency ratios share the current leftmost note as 1

String	Position / width	Actual	Target	Error, cents
1 · end	0	1	1	0 · exact
2 · inner	1/2	8/7	28/27	+168.21
3 · inner	2/3	6/5	16/15	+203.91

String	Position / width	Actual	Target	Error, cents
4 · end	1	4/3	4/3	0 · exact

The sound plays your four notes from left to right, pauses, then plays the four target notes. Both sequences use the same current leftmost pitch. The target ratios come from the Archytas report in I.13, retained here on pp.62–63. Sine oscillators make the ratios audible; they establish no ancient performance or physical instrument response.

The line fixes more than its ends

With a fourth between the two end notes, their sounding lengths are q and 3q/4. A straight bridge between the fixed end contacts leaves no independent height for the two inner contacts.

$$\text{Sounding length} = q(1 - x/4)$$

$$\text{Frequency / left note} = 1/(1 - x/4)$$

$$\text{Required } x = 4(1 - 1/\text{target ratio})$$

The original positions 1/2 and 2/3 necessarily give ratios 8/7 and 6/5. Shortening q raises every pitch by the same factor; interval ratios and errors stay fixed. Moving the two strings can change those ratios. Our exact ruler has 504 equal parts, chosen to represent all three targets' required positions without rounding. It is a modern numerical convenience.

Ptolemy's II.2 extension uses a longer construction and an octachord, with a common movable support. This is a reduced four-string analogy using a fourth as its end interval, rather than his full construction. His comparison on pp.100–101 also discusses the practical burden of moving strings and changing their separations. The model omits retensioning, coupling, support thickness and finger reach.

Greek printed letters and Wallis's Latin diagram labels

Greek	α	β	γ	δ	ε	ζ	η	θ	κ	λ	μ
Latin	A	B	C	D	E	F	G	H	K	L	M

Readings and transcription

Six excerpts from Wallis's 1682 edition, prepared against the retained images. Greek lowercase segment letters have their bars omitted. Line breaks, end-of-line hyphens, Latin ligatures and long s are normalized. Each reading identifies whether it is a sentence or a selected phrase. Wallis's Latin is an early modern translation. New English and contextual notes have not received independent philological review.

Start with a square

Ἐκτίθενται τετράγωνον, ὡς τὸ αβγδ.

Exponunt utique quadratum, ut ABCD.

They set out a square, such as ABCD.

Ptolemy describes a construction attributed to mathematicians. This is a selected complete sentence. The workshop uses Wallis's Latin diagram letters. Greek barred lowercase segment labels are normalized to ordinary lowercase here.

Ἐκτίθενται · *Exponunt*

They set out. A present-tense construction instruction. It introduces a proposed geometrical figure, rather than an excavated object.

τετράγωνον · *quadratum*

a square. The figure has equal sides and right angles. The later extension on p.99 allows a more general parallelogram.

Whole printed page 97.

Halve two different sides

διελόντες δίχα τὰς αβ καὶ βδ

bisectis AB & BD

having divided AB and BD in half

An excerpt of the next instruction, stopping before the named division points. The Greek participle διελόντες gives an action completed before joining the lines. Wallis uses a Latin ablative absolute. The E midpoint runs across the board; the F midpoint runs along a string. Neither instruction places G at the centre.

δίχα · *bisectis*

in two equal parts. The Latin puts the division into the participle ‘having been bisected’. In the diagram, AE = EB and BF = FD.

διελόντες · *bisectis*

having divided. Greek describes what the constructors do. Latin describes what has been done to AB and BD.

Whole printed page 97.

One board, many intervals

πάσας τὰς συμφωνίας, καὶ τὸν τόνον

Consonantiae omnes, & Tonus

all the consonances, and the tone

A selected noun phrase from the result, not a full sentence. Greek uses the accusative after ‘producing’; Wallis’s corresponding result makes the consonances and tone its grammatical subject. His spelling Consonantiae is normalized from the printed ligature. The list includes octave + fourth (8:3), octave + fifth (3:1) and two octaves (4:1). ‘All’ belongs to that theoretical list; it is not a claim about every musical interval or every culture’s consonance judgments.

συμφωνίας · *Consonantiae*

consonances. The named interval relations. In this model, a ratio of lengths becomes an inverse ratio of frequencies.

τόνον · *Tonus*

tone. Here the whole-tone interval 9:8, produced by lengths 9 and 8. It is a relation between pitches, not the absolute reference frequency.

Whole printed page 98.

Place the strings, then the bridge

τὰς χορδὰς παραλλήλους τε τῇ αγ, καὶ ἰσοτόνους ἀλλήλαις

chordas ipsi AC parallelas, atque inter se aequitonas

the strings parallel to AC and equal in pitch to one another

A noun-phrase excerpt from Ptolemy's instruction to stretch the strings in his extension of the instrument. The shared bridge follows this initial setup. ἰσοτόνους means equal in pitch; it does not establish equal physical tension by itself. Our uniform strings additionally hold material and tension fixed. The retuning bench is a reduced four-string analogy, not the full extended octachord drawn in the source.

παραλλήλους · *parallelas*

parallel. The strings keep a common direction. Their distances across the board can change independently of that direction.

ἰσοτόνους ἀλλήλαις · *inter se aequitonas*

equal in pitch to one another. An initial condition before dividing their sounding lengths. A real moved string would have to be checked again.

Whole printed page 99.

One common support

τὸ κοινὸν ἔχειν ὑπαγωγέα, καὶ ἕνα, καὶ ἐπὶ μίαν θέσιν

quod commune habeat subductorium, unicumque, & eodem situ positum

having one common movable support, in a single position

An excerpt of the advantage Ptolemy assigns to this method. The Greek infinitive phrase names the advantage; Wallis uses a ‘that’ clause. ὑπαγωγέα is the under-sliding support or bridge. The source then changes the common bridge’s position while preserving a genus. Our scaled bridge illustrates the similar-triangle principle with four chosen strings.

κοινὸν · *commune*

common. One support crosses all the strings. Moving a single crossing independently would require a different support.

μίαν θέσιν · *eodem situ*

one position. This concerns the bridge’s shared placement at a given setting. It does not promise that its placement never changes.

Whole printed page 100.

The practical price

τὰς χορδὰς ὅλας παραφέρειν

quod chordas integras transferre oporteat

moving the strings as wholes

A Greek infinitive excerpt from Ptolemy’s comparison of the two methods. Wallis’s selected clause expands it to ‘that it is necessary to move the whole strings’. Ptolemy also discusses changing string separations, sometimes by a large amount, in altering a tuning. The bench shows lateral geometry; it does not calculate finger reach, retensioning or practical playability.

ὅλας · *integras*

whole. The strings move across the board as wholes, rather than acquiring an independently moved contact under one string.

παραφέρειν · *transferre*

to move across. An infinitive describing the operation that this method requires. Our two inner string controls make that operation visible.

Whole printed page 101.

Whole printed evidence

The pages retain their figures, notes and critical apparatus. The source PDF has 189864593 bytes; SHA-256 862839d2cd52dfc62004d0f68da4d8381417e8be69e3901db0d5f73568931884. These eight complete JPEGs and all fonts and licenses were verified by byte length and SHA-256 before embedding.

ΚΛΑΥΔΙΟΥ ΠΤΟΛΕΜΑΙΟΥ

ΑΡΜΟΝΙΚΩΝ

ΒΙΒΛΙΑ Γ'.

37-9-32

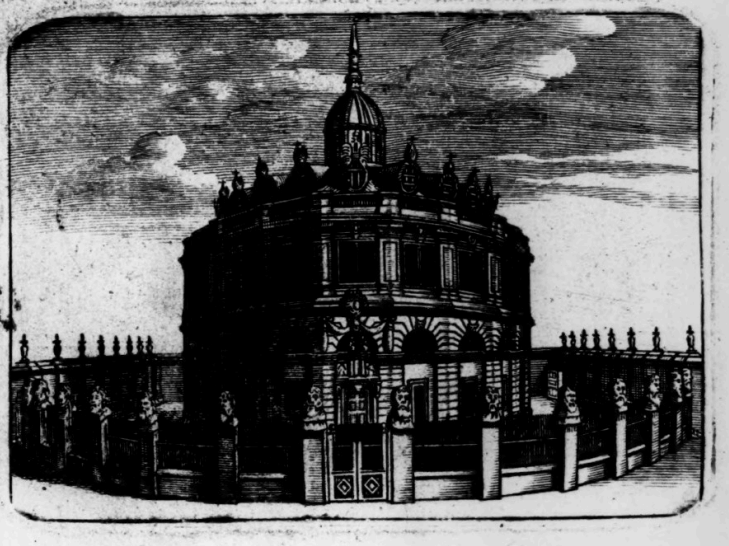
CLAUDII PTOLEMÆI

HARMONICORUM

LIBRI TRES.

Ex Codd. MSS. Undecim, nunc primum Græce editus.

JOHANNES WALLIS, SS. TH. D. Geometriæ Professor Savilianus *Oxoniæ*, Regiæ Societatis *Londini* Sodalis, Regiæque Majestati à Sacris; Recensuit, Edidit, Versione & Notis illustravit, & Auctarium adjecit.



OXONII,
E THEATRO SHELDONIANO, An. Dom. 1682.

καὶ παραβολῆς. Γίνοιτο δ' ἂν ἡ τοιαύτη τῆς ὀκτάχορδον κανόνα τοῦ διὰ πασῶν [c] λῆψις καθ' ἕτερον τρόπον, ἐπὶ τὸ καλούμενον ὄργανον Ἑλικῶνα, πεποιημένον τοῖς ἀπὸ τῶν μαθημάτων, εἰς τιὼ ἔνδειξιν τῶν ἐν ταῖς συμφωνίαις λόγων, ὑποτυπώσεως χάριν. Ἐκκείσθω τετράγωνον, ὡς τὸ ΑΒΓΔ. Καὶ, διελόντες δίχα τὰς ΑΒ [d] καὶ ΒΔ, τοιαύτη τὰ Ε καὶ Ζ [e] σημεῖα. Ζευγνύσθω μὲν τὰς ΑΖ καὶ ΖΗΓ· διαγέσθω δὲ, ἐπὶ μὲν τιὼ ΑΓ [f], διὰ μὲν τοῦ [g] Ε, τοῦ ΕΘΚ· διὰ δὲ τοῦ Η, [h] τοῦ ΛΗΜ. Αὐτόθεν μὲν οὖν, ἡ ΑΓ, ἑκατέρας [i] τῶν ΒΖ καὶ ΖΔ ὑπόκειται διπλασία· καὶ ἑπὶ τούτων ἑκατέρα, τοῦ ΕΘ· ἐπεὶ καὶ ἡ ΑΒ τοῦ ΑΕ. Ὡς τε καὶ τοῦ ΑΓ, τοῦ μὲν ΕΘ, τετραπλασίαν εἶναι· λοιπὴ δὲ τοῦ [k] ΘΚ ὑπερέχον. Δείκνυται δὲ ὅτι καὶ ἡ ΜΗ τοῦ [l] ΛΗ διπλασία ἐστὶν· ἐπιζευχθεισῶν γὰρ ὡς μὲν ἡ ΔΓ πρὸς τιὼ ΓΜ, οὕτως ἡ ΔΒ πρὸς τιὼ ΗΜ· ὡς δὲ ἡ ΒΑ πρὸς τιὼ ΑΛ, [m] τουτέστι πάλιν ὡς ἡ ΔΓ πρὸς τιὼ ΓΜ, οὕτως ἡ ΒΖ πρὸς τιὼ ΛΗ· καὶ διὰ τοῦτο, ὡς ἡ ΒΔ πρὸς τιὼ ΗΜ, [n] οὕτως ἡ ΒΖ πρὸς τιὼ [o] ΛΗ· καὶ ἐναλλὰξ, ὡς ἡ ΒΔ πρὸς τιὼ ΒΖ, οὕτως ἡ ΜΗ πρὸς τιὼ ΛΗ. Γίνεται ἄρα καὶ ἡ ΑΓ τοῦ μὲν ΗΜ ἡμιολία· τοῦ δὲ ΗΛ τετρα-

Fieri autem poteſt ipſius Diapaſon applicatio ad Octachordum canonem alio adhuc modo; ſecundum Inſtrumentum Helicona dictum; (a Mathematicis conſtructum, ad exhibendas Conſonantiarum rationes;) quaſi ad hunc modum. Exponunt utique quadratum, ut ABCD. Atque, biſectis AB & BD, in E & F, jungunt AF & BGC rectas; rectæque AC, per E punctum, parallelam agunt EHK; & per punctum G, parallelam LGM. Hinc utique conficitur, AC, utriuſque ipſarum BF & FD, dupla: atque harum utraque, dupla ipſius EH; (propter AB *duplam* ipſius AE:) Adeoque AC, ipſius EH quadrupla; reliquæ autem HK ſeſquitertia. Demonſtratur autem & MG ipſius LG dupla: (quoniam ut DC ad CM, ſic eſt DB ad GM; ut autem BA ad AL, quod eſt iterum ut D ad CM, ſic BF ad LG: & propterea, ut BD ad GM, ſic BF ad LG; &, alternatim, ut BD ad BF, ſic MG ad LG.) Eſt igitur AC ipſius GM ſeſquialtera; ipſiuſque GL tripla. Diſpoſitis itaque quatuor chordis æquitonis,

in

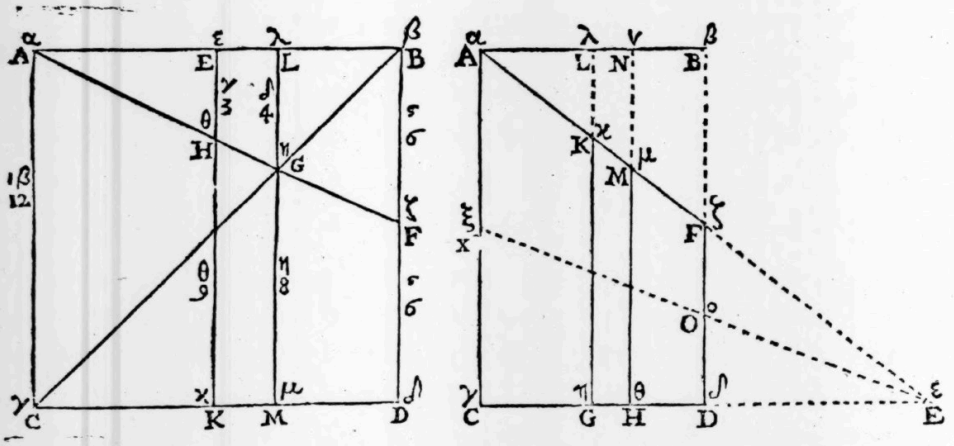
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dunt; (hoc eſt, in eadem longitudine, eundem ſonum:) differunt autem pro varia longitudine; ut nempe ſonorum acumina ſint in longitudinum ratione reciproca.

c AB. κείσις, & C. in margine; ſed, in textu χῆσις. *d* AB. addunt πλευρὰς. *e* AB. addunt σημεῖα. *f* AB. addunt εὐθεῖαν. *g* AB. deeſt ε τ. *h* AB. deeſt τ. *i* B. πδ, in marg. τ, A. τ in textu. *k* GI. θκ. *l* CK. κλ. *m* CK. deeſt, τουτέστι πάλιν ὡς ἡ ΔΓ, πρὸς τὴν ΓΜ. *n* AB. ἕτως ἡ ΒΖ. *o* GI. λΒ.

in his ipſis AC, EK, LM, & BD rectarum poſitionibus: ſuppoſitoque eiſdem Canonio in ſitu rectæ FGHA: numeriſque apte accommodatis; rectæ nimirum AC, 12; HK, 9; GM, 8; utrique BF & FD, 6; itemque LG, 4; & EH, 3: Conficientur Conſonantiæ omnes, & Tonus. Nempe, Dia-teſſaron, (ut quæ eſt in ratione ſeſquitertia,) ab AC & HK; item a GM & FD; atque ab LG & EH: Dia-pente vero, (utpote in ratione ſeſquialtera,) ab AC & GM; item, ab HK & FD; atque a BF & LG: Dia-paſon vero, (in ratione dupla,) ab AC & FD; item a GM & LG; atq; a BF & HE: Dia-paſon & dia-teſſaron, (in ratione 8 ad 3,) a GM & HE: Dia-paſon &

σία. Ὡς τε διαζαθφσῶν χορδῶν τεσσάρων ἰσοτόνων καὶ τὰς αὐτὰς θέσεις, [p] τὰς τῶν ᾱγ, καὶ ᾱκ, καὶ λμ, καὶ βδ εὐθφῶν· καὶ ὑπαχθέντος αὐτοῖς κανονίου καὶ [q] τοῦ τὴν ζηθα θέσιν· ἐφαρμοσθέντων τε ἀειθωῶς, τῇ [r] μ̄ ᾱγ τοῦ τφς ῑϛ, τῇ δὲ ϝκ τοῦ τ̄ ϝ, τῇ δὲ η μ τοῦ τ̄ η, ἑκατέρα δὲ τ̄ βζ καὶ ζδ τοῦ τ̄ ϛ̄, καὶ πάλιν τῇ μ̄ λη τοῦ τ̄ δ, τῇ δὲ εθ ϛ̄ τ̄ γ̄· ἀποτελεσθῆς πάσας τὰς συμφωνίας, καὶ τὸν τόνον· τὴν μὲν διὰ τεσσάρων [s] καὶ καθ' ὑπεροχὴν λόγον συνιστα μένης, ὑπό τε τ̄ ᾱγ καὶ ϝκ, καὶ ὑπὸ τ̄ ημ καὶ ζδ, καὶ ὑπὸ τ̄ λη καὶ [t] εθ· τὴν δὲ διὰ πέντε καὶ ἐν τῷ ἡμιολίῳ λόγῳ, ὑπό τε τφς ᾱγ καὶ ημ, καὶ ὑπὸ τφς ϝκ καὶ ζδ, καὶ ὑπὸ τ̄ βζ καὶ λη· τὴν δὲ διὰ πασῶν καὶ καθ' ὑπεροχὴν διπλάσιον λόγον, ὑπό τε [u] τ̄ ᾱγ καὶ ζδ, καὶ ὑπὸ τ̄ ημ καὶ λη, καὶ ὑπὸ τ̄ βζ καὶ θε· τὴν δὲ διὰ πασῶν καὶ διὰ τεσσάρων, ἐν τῷ τ̄ [x] η πρὸς τὸν γ λόγῳ, ὑπό τ̄ ημ καὶ θε· τὴν



p AB. ταῖς. q K. τ̄ τ̄, ABC. τ̄ αθηζ. r GI. deeſt μ̄. ſ AB. deeſt καὶ. t GI. deeſt τ̄. u GI. τ̄. x K. γη πρὸς τ̄ τὸ γ η, AB. πρὸς τὸ γ.

y B.

Wallis, 1682 · printed p.98 · PDF p.120 · complete page

δὲ διὰ πασῶν ϗ διὰ πέντε [y] ϗ κατὰ ⟨Ⓜ⟩ τριπλάσιον λόγον, ὑπὸ τὴν ΑΓ [z] ϗ ΛΗ, [a] ϗ ΘΚ ϗ ΦΕ. τὴν δὲ δὶς διὰ πασῶν, ϗ κατὰ τὸ τετραπλάσιον λόγον, ὑπὸ τὴν ΑΓ καὶ ΕΘ. καὶ ἔτι τὸν τόνου, κατὰ ⟨Ⓜ⟩ ἐπόγδοον λόγον, ὑπὸ τὴν [b] ΦΚ καὶ ΗΜ. Παρὰ δὴ τοῦτο τὸ ὄργανον, ἐὰν ἐκθώμεθα παραλληλόγραμμον ἁπλῶς, ὡς τὸ ΑΒΓΔ. ϗ νοήσωμεν ἐπὶ μὲ ΑΒ ϗ ΓΔ κεῖσθαι τὰ ἀποφάλματα τῶν χορδῶν· τὰς δὲ ΑΓ καὶ ΒΔ κατὰ τοὺς ἄκρους φθόγγους ἑκατέρους διὰ πασῶν· ἔπειτα, προσεκβαλόντες τὴν ΓΔ ἴσῃ [c] τῷ ΔΕ, καταλέμωμεν, ἀντὶ τῶν χαμονίων, τῷ ΓΔ πλευρᾷ τοῖς [d] οἰκείοις τῶν γενῶν λόγοις, ὑπὸ ἑκάτερα τὸ ὀξὺ πέρας [e] ἀποτιθέμενοι· καὶ διὰ τῶν γινομένων ἐπ' αὐτῆς τομῶν [f] τείνωμεν τὰς χορδὰς παραλλήλους τε τῇ ΑΓ, ϗ ἰσοτόνους ἀλλήλαις· ϗ, τοῦτο γινομένου, τὸ κοινὸν ἐσόμενον ὑπαγωγεὺς τῶν χορδῶν, [g] ὑποβάλωμεν [h] αὐταῖς, καὶ [i] τῷ τὸ ὑπεζευγνύσης τὰ Α Ε σημεῖα θέσιν, ὡς τὴν ΑΖΕ. Ποιήσωμεν πάλιν [k] τὰ μήκη τῶν χορδῶν, ἐν τοῖς αὐτοῖς λόγοις· ὡς τε ἐπιδέχεσθαι τῷ τῶν ἐφηρμοσμένων τοῖς γένεσι [l] λόγων ἀνάκεισιν. Ἐπεὶδήπερ ὡς [m] αἱ ἀπὸ ΔΕ λαμβα-

dia-pente, (in ratione tripla,) ab AC & LG; itemque ab HK & HE; Diſ-dia-paſon vero, (in ratione quadrupla,) ab AC & EH: Tonus denique, (in ratione ſeſquioctava,) ab HK & GM. Porro; ſecundum hoc ipſum Inſtrumentum, ſi exponamus utcunque Parallelogrammum, ut ABCD; (*ſive quadratum ſit, ſive aliud utcunque parallelogrammum*;) atque intelligamus AB & CD, ad Chordarum Apopſalmata *ſeu Finitores* (*quibus chordarum longitudines utrinque terminantur*;) & AC BD, ad extremos ſonos ipſius Dia-paſon: Deinde, (protracta, ipſi CD, æquali DE,) ſecemus (pro canoniis) CD latus, pro rationibus cuique generum propriis; poſito, verſus E, extremo acutiori: perque hujus ſectiones ſic factas, tendamus chordas ipſi AC parallelas, atque inter ſe æquitonas: atque, hoc demum facto, commune futurum chordis ſubductorium, eiſdem ſubjiciamus, in ſitu rectæ quæ puncta A E conjungit; ut AFE: Conficiemus iterum chordarum longitudines, in eiſdem rationibus; quæ itaque recipient rationum examen ſingulis generibus congruentium. Quoniam, ut ſe habent

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y B. deeſt ϗ. *z* A. deeſt, ϗ λη, ϗ θκ, ϗ φε· τὴν δὲ δὶς διὰ πασῶν, ϗ κατὰ τετραπλάσιον λόγον ὑπὸ τὴν ΑΓ. *a* BCK. deeſt ſaltem ϗ θκ ϗ φε. *b* GI. τὴν ΦΗ. *c* GI. τῇ. *d* AB. ἄκροις, C. οἰκίοις, & ſimiliter aliquoties. *e* CK. ὑποτιθ--. *f* AB. ὑποτείν--. *g* C. ὑποβαλλ--. *h* K. αὐτὰς. *i* CK. τῷ ὑποζευγνύσ. *k* GI. πάντα. *l* C. λόγον ἀνάχρησιν. *m* AB. δὶ.

bent inter ſe, rectæ (ab E ſumptæ) in CD; ita ſe habebunt inter ſe, quæ ab earum terminis uſque ad AF, aguntur ipſi AC parallelæ. Puta, ut EC ad ED, ſic CA ad DF: quæ propterea Dia-paſon exhibebunt, quoniam in ratione dupla. Si autem porro, a CD, auferamus CG, rectæ CE partem quartam; & CH, ejuſdem partem tertiam: atq; ad G & H chordas ſtatuamus, ut GKL, & HMN, primitus poſitis æquitonas; adeoque fiat AC, ipſius GK, ſeſquitertia; ipſiuſque HM, ſeſquialtera; itemque, ipſius DF, ſeſquitertia HM, & GK ſeſquialtera; itidemque GK ipſius HM ſeſquioctava: Facient & hæ ad invicem, conſonantias rationibus illis conſentaneas. Simileq; pariter conſequetur in ſumptis ſectionibus binorum tetrachordorum intermediis, ſecundum debitas examinatorum rationes. Prior autem ille modus, ad hunc comparatus, hoc habet magis expeditum; quod non opus habeat chordarum inter ſe diſtantias mutare: Hic autem, præ illo; quod commune habeat ſubductorium, unicumque, & eodem ſitu poſitum; Et porro quod poſſit, illud *ſubductorium* (per E tranſiens) detruſum (puta in ſitum

νόμεναι κατὰ τὴν γ δ πρὸς ἀλλήλας ἔχουσιν, [n] ὅπως καὶ αἱ διὰ τῶν πρώτων αὐτῶν ἀναγόμεναι [o] πρὸς τὴν αγ μέρει τ α ζ, ἐξουσι πρὸς ἀλλήλας. Οἷον, ὡς ἡ εγ πρὸς τὴν ε δ, ὅπως ἡ γα πρὸς τὴν δ ζ. [p] διόπερ αὐταὶ μὲ ποιήσουσι τὸ διὰ πασῶν, ὅπι διπλάσιος αὐτῶν ὁ λόγος. Ἐαν δὲ ἀπολαβόντες πάλιν, ἀπὸ τῆς γ δ, τὴν μὲν γ η, κατὰ τὸ τέταρτον μέρος τ εγ· τὴν δὲ γ θ κατὰ τὸ τρίτον τ αὐτῆς· [q] ἀναστήσωμεν καὶ διὰ τῶν η καὶ θ χορδὰς, ὡς τὰς η κ λ καὶ θ μ ν, ταῖς προτάσαις ἰσοτόνους· ὡς τε καὶ τὴν [r] τε αγ τ μὲ ηκ γίνεσθαι ὀκτάτονον, τ δὲ θ μ ἡμιολίαν· καὶ πάλιν τ δ ζ τ [f] μὲ θ μ ὀκτάτονον, [t] τὴν δὲ ηκ ἡμιολίαν· καὶ ἔτι τὴν [u] ηκ τ θ μ ἐπόγδοον· ποιήσουσι καὶ αὐταὶ πρὸς ἀλλήλας τὰς [x] ἀκολούθως τοῖς λόγοις συμφωνίας· Τῷ προσπλησίον [y] προσακολουθήσαι καὶ ὅπι τῶν μεταξὺ [z] τοῖν τετραχόρδοιν [a] λαμβανομένων τμημάτων ἐν τοῖς οἰκείοις τῶν ἀνακειμένων λόγοις. Ἐχει δ' ὁ μὲ προτέτ πρόπος πρὸς τοῦτον προχειρότερον, [b] ὃ μὴ δεῖν κινεῖν τὰς ἀπ' ἀλλήλων διαστάσεις τῶν χορδῶν· ὅτι δὲ πρὸς ὀξύτερον, τό τε κοινὸν ἔχειν ὑπαγωγέα, καὶ ἕνα, καὶ κατὰ μίαν θέσιν· καὶ ἔτι, ὃ δύναται καὶ ὑποβαζόμενον αὐτὸν διὰ τοῦ ε, ὡς ὅπι τὴν ξ ο ε θέσιν, ὀξύτερον ποιεῖν ὅλον τὸν

n AB. ὅπως. *o* GI. πρὸς. *p* AB. διόπ προ. *q* AB. --ήσομεν. *r* GI. deeſt τε. *ſ* ABC. deeſt μὲ. *t* AB. τὴ. *u* GI. μὲ νηκ. *x* C. --θως, in reliquis --θως. *y* C. --σονλ. item in AB, ſed in his reſtituitur --σαυλ. *z* GI. τὴν τετραχόρδων, in reliquis --οιν. *a* GI. marg. προλαμβ--. *b* GI. πὶ, AB. πὴ.

c AB.

τόνον, μενόσης δ καὶ [c] τὸ γένος ἰδιότητος. Επεὶ γὰρ ὡς ἡ γα, φέρε εἰπεῖν, πρὸς τὴν [d] ζοδ, οὕτως ἐστὶν ἡ ξγ πρὸς τὴν οδ· καὶ ἐπὶ τῶν ἄλλων [e] ὁμοίως. Πάλιν τ' αὖ κατασκελέστερον ὁ μὲν πρότερος ἔχει τρόπος πρὸς τοῦτον, [f] τὸ πλέονα λῆξιν κινῆν ὑπαγωγίδια καθ' ἑκάστην ἁρμογήν· ὅτος δὲ πρὸς ἐκεῖνον, ὃ τὰς χορδὰς ὅλας παραφέρειν· καὶ μηκέτι κατ' ἴσας αὐτῶν διαστάσεις, ἀλλὰ πολλαχῆ μακρῷ διαφερούσας, συντελεῖν τὰς τῶν [g] ἐπιφαύσεων μεταβάσεις.

XOE,) universum tonum acutiorem reddere; manente sua generis proprietate; (Quoniam est, verbi gratia, ut CA ad FOD, sic XC ad OD; & similiter in reliquis.) Iterumque, hoc habet operosius prior ille modus, præ hoc altero, quod plura oporteat removere subductoria, in singulis aptationibus: hic autem præ illo, quod chordas integras transferre oporteat; atque etiam quod chordarum distantias (non æquales quidem, sed) sæpenumero multum differentes, transire oporteat, tactus expediendo.

γ'.

Περὶ τῶν ἐν ταῖς πρώταις συμφωνίαις εἰδῶν.

CAP. III.

De eis quæ sunt, in primis Consonantiis, Speciebus.

ΤΑ μὲν οὖν περὶ τὰς συμφωνίας, καὶ τὰς [a] ἐμμελείας τῶν πρὸς ὑπόφαλμα κειμένων φθόγγων θεωρούμενα, μέχρι [b] τοσούτων ἡμῖν [c] ἐκτεθείσθω· συνεκλαμβανομένων τοῖς συμφώνοις καὶ τῶν ὁμοφωνιῶν. Συνεχῶς δὲ τούτοις ὅλως ἂν περὶ τῶν συστημάτων λόγος προθεωρητέον, τὰς, πρὸς τὸ καλούμενον εἶδος, τῶν πρώτων συμφωνιῶν διαφορὰς, ἐχούσας ὅτως. Εἶδος

DE Consonantiis igitur & Concinnitatibus, sonorum ad Apopsalma positorum, quæ considerandæ erant, hactenus esto nobis explicatum; Unisonantiis juxta cum Consonantiis comprehensis. Cumque jam proxima esset de Systematis dissertatio; definiendæ prius veniunt primarum Consonantiarum illæ differentiæ quæ sunt (ut loquuntur) secundum Speciem, quæ quidem sic se habent. Species

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c AB. deest τὸ. *d* AB. ζδ. *e* AB. ὁμοίων. *f* AB. deest τὸ. *g* AB. ὑπὸ τάσεων.
CAP. 3. *a* C. --λίας, & sic sæpe. *b* AB. ποσῶν. *c* GI. ἐκτετ--.

ceſſuum Commenſuratio.) Cum vero hunc ſibi omnino ſcopum propoſuerit; ab eo tamen in quibuſdam errare plane deprehenditur: In pleriſque autem, eidem adhæret; ita tamen ut manifeſte recedat ab eis quæ ſenſibus directe ſunt comperta: prout mox videbimus in tetrachordorum, ſecundum illum, diviſione. Tria nimirum conſtituit ille Genera; Enarmonium, Chromaticum, & Diatonicum. Singulorum autem diviſionem ad hunc modum inſtituit. Rationem ſequentem, in tribus his Generibus conſtituit eandem; nempe, ſeſquivigeſimamſeptimam: Mediam vero, in Enarmonio, ſeſquitrigeſimamquintam: in Diatonico, ſeſquiſeptimam. Unde colligitur; Generis Enarmonici, præcedentem eſſe ſeſquiquartam; Diatonici vero, ſeſquioctavam. Sed, in genere Chromatico, ſonum ab acutiſſimo ſecundum, ab eo deſumit qui eundem ſitum habet in Diatonico: Ait enim, ſecundum ab acutiſſimo in genere Chromatico, ad ſimilem in Diatonico, eam habere rationem quam 256 ad 243. Conſtituuntur itaque hæc tria tetrachorda,

τῶν ἐμμελῶν ὄντων [r] συμμέτρων τὸν ὑποκείμενον. Ταύτην δ' ὅμως τὴν προθέσεως ἀναπόδραστον, εἰς [s] ἕνια μὲν καὶ [t] τέλεον αὐτῆς φαίνεται διαμαρτάνων· ἐν δὲ τοῖς πλείστοις, τοῦ μὲν τοιούτου παρίσταται, ἀπάδων δὲ σαφῶς τῶν αἰσθήσεων ἤδη ταῖς αἰσθήσεσιν ὁμολογουμένων· ὡς [u] αὐτίκα εἰσόμεθα ἐκ τῆς [x] κατ' αὐτὸν τῶν τετραχόρδων διαιρέσεως. [y] Τρία μὲν [z] τοίνυν ὑπέθετο [a] ὑφήσεσι γένη· τό τε ἐναρμόνιον, καὶ τὸ χρωματικόν, καὶ [b] τὸ διατονικόν· Ἑκάστου [c] δὲ αὐτῶν ποιεῖται τὴν διαίρεσιν, [d] ὧδε. Τὸν μὲν γὰρ ἐπόμενον λόγον, ὑπὲρ τὰ τρία γένη, Ⓜ αὐτὸν [e] ὑφήσει, καὶ [f] ὑπερεικοσιεβδόμου· Τὸν δὲ μέσον, ὑπὲρ μὲν τὸ ἐναρμόνιον, ὑπερτριακοστόπεμπτον· ὑπὲρ δὲ τὸ διατονικόν, ὑπερέβδομον· Ὥστε καὶ τὸν ἡγούμενον τῶν μὲν ἐναρμονίου γένους συνάγεσθαι ὑπερτέταρτον· τῶν δὲ διατονικοῦ, ἐπόγδοον. [g] Τὸν δὲ, ἐν ᾧ τὸ χρωματικῷ γένει, δεύτερον ἀπὸ τοῦ ὀξυτάτου [h] φθόγγου λαμβάνει, διὰ τὸ τὴν αὐτὴν θέσιν ἔχοντος ἐν ᾧ διατονικῷ· φησὶ γὰρ, λόγον ἔχειν [i] τὸν ἐν ᾧ χρωματικῷ δεύτερον ἀπὸ τῶν ὀξυτάτου, πρὸς τὸν ὅμοιον [k] τὸν ἐν τῷ διατονικῷ, [l] τῶν σνϛ πρὸς τὰ σμγ. [m] Συνίσταται δὴ τὰ τοιαῦτα

[n] τρία

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r Συμμέτρων. Ita ſcilicet (quippe hoc vult) ut vel comparatorum alterum (ut in Multiplicibus) vel ſaltem eorum Differentia (ut in Superparticularibus) ſit utriuſque communis menſura. ſ AB. ἕνα. t K. τέλειον. u S. αὐτα. x AB. κατ' αὐτῆ. CK. κατ' αὐτὸν. GI. κατ' αὐτῶν. S. κατ'. y AB. τρίτος. z GI. ἓν ἤδη. a ABS. ὑφίσατω. b S. deeſt τὸ. c GI. δ'. d C. ὅπω. e ABS. ὑφίσατω. f S ἐικοστεβ--. g GI. τῶν. h AB. φθόγγω. i CS. deeſt τ. k GI. τῶν, reliqui τ. l S. τ. m CK. συνίστατι.

n S.

[n] τρία τετράχορδα, καὶ τὰς ἐκκειμένας λόγους, ἐν [o] τοῖς πρώτοις ἀριθμοῖς τούτοις. [p] Ἐὰν γὰρ τὰς μὲν ὀξυτάτας τῶν τετραχόρδων ὑποστησώμεθα ρφιβ, τοὺς δὲ βαρυτάτους καὶ τὸν ὑπερτέλειον λόγον [q] τῶν αὐτῶν βιγ· ἔσται μὲν ποιήσαι τὸν [r] ὑπτακοσιοστέβδομον [s] πρὸς τὸν [t] ρτμδ· καὶ τοσούτων ἔσονται πάλιν ἐν τοῖς τρισὶ γένεσιν οἱ δευτέροις ἀπὸ τῶν βαρυτάτων· τῶν δ' ἀπὸ θατέρου ὀξυτάτου δευτέρων, [u] ὁ μὲν τοῦ ἐναρμονίου γένους [x] ἔσται ρωζ· ἔσονται γὰρ πρὸς μὲν τὸν ρτμδ ποιοῦντες τὸν ὑπτακοσιοστόπεμπτον λόγον· πρὸς [y] δὲ τὸν ρφιβ τὸν ὑπτέταρτον· ὁ δὲ τοῦ διατονικοῦ γένους, τῶν αὐτῶν ἔσται ρψα· καὶ ταῦτα γὰρ πρὸς μὲν τὰ ρτμδ τὸν ὑπτέβδομον ποιοῦντες λόγον· πρὸς δὲ τὰ ρφιβ τὸν ἐπόγδοον· ὁ δὲ τοῦ χρωματικοῦ, καὶ αὐτὸς ἔσται τῶν αὐτῶν

secundum rationes expositas his numeris minimis. Siquidem sonos tetrachordorum acutissimos constituamus 1512; adeoque gravissimos (in ratione nimirum sesquitertia) eorundem 2016: Facient hæc rationem sesquivigesimamseptimam, ad 1944: totidemque erunt, in tribus generibus, secundi a gravissimis. Secundorum autem ab acutissimo; est quidem ille generis Enarmonii, 1890; (quippe hæc ad 1944 rationem faciunt sesquitrigesimamquintam; ad 1512 autem, sesquiquartam:) Generis vero Diatonici, eorundem 1701, (nam hæc ad 1944 rationem faciunt sesquiseptimam; ad 1512, sesquioctavam:) Chromatici autem generis, eorundem erit 1792; (quippe

	Εναρμόνιον.		Χρωματικόν.		Διατονικόν.		
Εἰς ὀξύ, ὑπάτμημος.	ρφιβ	1512	ρφιβ	1512	ρφιβ	1512	*In Acutum, præcedens.*
	ἐπὶ δ'	$1\frac{1}{4}$		$1\frac{1}{27}$	ὅπ η'	$1\frac{1}{8}$	
	ρωζ	1890	ρψβ	1792	ρψα	1701	
	ὅπ λε'	$1\frac{1}{35}$		$1\frac{9}{224}$	ὅπ ζ'	$1\frac{1}{7}$	
	ρτμδ	1944	ρτμδ	1944	ρτμδ	1944	
	ὅπ κζ'	$1\frac{1}{27}$	ὅπ κζ'	$1\frac{1}{27}$	ὅπ κζ'	$1\frac{1}{27}$	
Εἰς βαρύ, ἑπτόμμος.	βις	2016	βις	2016	βις	2016	*In Grave, consequens.*
	Enarmonium.		*Chromaticum.*		*Diatonicum.*		
	$\frac{5}{4} \times \frac{36}{35} \times \frac{28}{27} = \frac{4}{3}$		$\frac{32}{27} \times \frac{243}{224} \frac{28}{27} = \frac{4}{3}$		$\frac{9}{8} \times \frac{8}{7} \times \frac{28}{27} = \frac{4}{3}$		

n S. deest τρία. o GI. πρώτοις τοῖς. p S. καὶν. q AB. deest τῶν αὐτῶν. r CS. ὑπτακοσούδομον. Et sic aliquoties υ pro β. Sic Hispani v & b promiscue sæpe & scribunt & proferunt: (sed & ex nostris multi ὰ & ων, quasi ev & av.) s AB. deest τὸν. KS. πρὸς τὸν. t GS. quasi perpetuo π pro ρ. u S. ὅμως. x GI. ὀξὺν. y S. deest δὲ.

z S.

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